

Automated digital transformation for pedestrian suspension bridges using hybrid semantic structure from motion

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ABSTRACT

Digital transformation is employed to create digital models that reflect the current state of infrastructure. Conventional semantic structure from motion methods effectively generated digital models of bridges segmented by components through semantic segmentation. However, these methods encounter significant challenges in the digital transformation of pedestrian suspension bridges: the inaccurate modeling of cables due to the inherent limitations of raster data in representing cables. To address these issues, this paper proposes a hybrid semantic structure from motion framework for pedestrian suspension bridges by integrating segmented cables in vector format with other raster format data. The performance of the proposed system was validated through full-scale experiments on a pedestrian suspension bridge in South Korea, achieving a 26.49% improvement in accuracy compared to conventional methods. By automating key aspects of digital transformation, the proposed framework is expected to provide solutions for the management of bridge infrastructure, enhancing safety and operational resilience.

1. Introduction

Pedestrian suspension bridges are gaining popularity due to their significant appeal to tourists. These pedestrian suspension bridges are often installed in scenic and risky locations, such as picturesque rivers or gorges, to maximize tourism effects, and their lengths are competitively increasing [1,2]. However, the long spans and hazardous installation locations chosen to maximize tourism appeal can lead to severe consequences in the event of accidents. In 2017, the Carrizal River suspension bridge near the city of Tosagua, Ecuador, collapsed, causing 32 injuries [3]. In 2022, the cable of a pedestrian suspension bridge in Gujarat, India, snapped, resulting in an accident that caused 141 fatalities [4]. In 2020, a lightning rod fell from the main tower onto the Nakgang suspension bridge in Gyeongsangbuk-do, South Korea [5]. As the number of pedestrian suspension bridges has increased, many accidents have occurred, and the increasing lengths and hazardous installation locations of these bridges have further intensified the severity of these incidents.

To prevent accidents related to pedestrian suspension bridges, the need for continuous monitoring of these structures is becoming increasingly critical. For the safety inspection of pedestrian suspension bridges, key structural components, such as the pylon, deck, and cable, need to be inspected [6]. In particular, the sag of the cables, defined as the vertical intervals of the main cable in the main span, is associated with

the stability of the bridge and cable tensile forces [7]. Therefore, monitoring this parameter is crucial for ensuring the safety and maintenance of pedestrian suspension bridges. However, current safety inspections for most pedestrian suspension bridges are primarily conducted through visual assessments by experts, which have inherent limitations. Manual inspection is resource-intensive and inefficient. Moreover, pedestrian access to the suspension bridge must be restricted during the inspection process, and safety incidents may occur during inspections. Finally, since these inspections rely on visual assessments by experts, the results can be significantly influenced by human subjectivity.

Recently, digital models of structures have been increasingly employed to enhance the efficiency of structural maintenance. By creating digital representations of structural components, not only the design and construction processes but also maintenance can be carried out more effectively [8]. Contrary to the numerous challenges associated with manual inspections, the use of digital models for structural safety inspections facilitates easy verification of component details. This approach obviates the need for on-site expert evaluations, thereby enhancing both efficiency and safety. Furthermore, it allows for quantitative assessments of results, significantly reducing the susceptibility to human subjectivity. As a result, the global adoption of digital models for structures has been rapidly increasing [9–12]. However, structures of smaller scale or older constructions, such as pedestrian suspension

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bridges, are less likely to have been constructed with digital models during their construction phase compared with larger structural projects. Additionally, for these structures, the creation of accurate digital models is complicated due to the loss of original design drawings. Even when design drawings are available, significant discrepancies between the actual structure and the drawings can occur due to construction errors and deformations over time, which can hinder the development of accurate digital models.

To address these issues, digital transformation technologies using LiDAR or photogrammetry are being developed to scan existing structures and convert them into digital models [13–18]. Kim et al. [19] proposed a deep learning-based methodology for automated bridge component recognition using 3D point clouds, capable of processing complex structures and incorporating background elements to improve accuracy and reduce preprocessing efforts. Narazaki et al. [20] proposed an unsupervised domain adaptation (UDA) approach using the DAFormer framework to enhance deep learning-based visual recognition for bridge inspection, demonstrating significant performance improvements by integrating synthetic and real-world data to reduce reliance on extensive manual annotations. Golparvar-Fard et al. [21] proposed an algorithm for semantic segmentation and recognition of highway assets from video frames, using car-mounted cameras to create detailed 3D models and accurately identify various assets. Lee et al. [22] obtained point clouds of bridges using LiDAR sensors and automatically extracted design parameters. Although these laser scanner-based digital transformation technologies can accurately generate 3D point cloud models of structures, they require traffic control during measurements and are time-consuming and costly due to the need for laser scanning sensors. To address these limitations, Park et al. [23] proposed a deep learning-based semantic structure from motion (SSfM), which constructs 3D digital models by segmenting structural components using images captured by unmanned aerial vehicles (UAVs) for railroad bridges. T. Czerniawski et al. [24] proposed a method called Density Based Spatial Clustering of Applications with Noise (DBSCAN) for detecting all planar objects within building point clouds captured by consumer-grade depth cameras. Furthermore, Kim et al. [25] proposed a framework for automated bridge component recognition based on close-range images. These methods, which utilize cameras to reduce cost and time, have rapidly expanded with the advancement of UAV technology. However, most digital transformation technologies are focused on buildings, road bridges and railroad bridges, with limited research conducted on pedestrian suspension bridges.

Cables are the most critical structural components of pedestrian suspension bridges. Unlike other types of bridges, pedestrian suspension bridges rely on cables to support all loads, making the stability of the main cables critical [6]. Therefore, accurate modeling of cables is essential when creating digital models of pedestrian suspension bridges. However, applying the current SSfM technology to the digital transformation of pedestrian suspension bridges presents a significant issue: inaccurate modeling of cables. The existing SSfM method utilizes semantic segmentation to represent structural component information using images of the structure. These images are represented as raster data, where the resolution is determined by the pixel grid spacing. The cables in pedestrian suspension bridges are very thin, with thicknesses ranging from a few pixels to less than a single pixel in the image. Representing such thin components in raster format results in substantial inaccuracies.

This article presents an automated digital transformation framework that achieves effective digital modeling of pedestrian suspension bridges by representing cables in vector format through a hybrid semantic structure from motion based on semantic segmentation and wireframe parsing. The study is conducted in three main phases: (1) deck and pylon semantic segmentation, (2) cable wireframe parsing, and (3) 3D reconstruction and digital model integration. In Phase 1, the structural components of the pedestrian suspension bridge, specifically the larger components such as the deck and pylon, are represented in raster format through semantic segmentation. In Phase 2,

the smaller components, such as the cables, are represented in vector format through wireframe parsing. Finally, in Phase 3, the semantic information obtained in raster and vector formats from Phases 1 and 2 is transformed into 3D using structure from motion and 3D transformation techniques, and integrated into a unified digital model. This process allows for the automatic generation of a 3D digital model of the pedestrian suspension bridge, segmented by components, and enables accurate measurement of cable sag. The contributions of this paper are as follows:

- Segmentation of cables in the 3D domain using a vector format: Recognizing the difficulty of representing these thin structural components with raster format, a semantic segmentation approach based on wireframe parsing is proposed.
- Hybrid Semantic Structure from Motion (SSfM) technology: This leverages both semantic segmentation for raster data and wireframe parsing for vector data, facilitating the 3D digital transformation of structures encompassing both data formats.
- Segmentation of cables in the 3D domain using a vector format: Recognizing the difficulty of representing these thin structural components with raster format, a semantic segmentation approach based on wireframe parsing is proposed. The test results showed a 8.2% reduction in the relative error of sag ratio predictions and a 26.49% increase in the accuracy of SSfM results compared to conventional methods.
- This study contributes to information technologies in maintenance by advancing automated inspection and maintenance processes for pedestrian suspension bridges.

2. Background

This study introduces a hybrid SSfM technology that combines semantic segmentation (for raster data) and wireframe parsing (for vector data) to enable the 3D digital transformation of structures using both raster and vector formats. The following sections provide detailed background information: (2.1) Raster and vector data formats, (2.2) Conventional semantic segmentation techniques for raster data, (2.3) Wireframe parsing for vector data, and (2.4) 3D point cloud reconstruction using structure from motion.

2.1. Raster and vector data

Raster data is a data format that represents objects in a grid pattern, with pixels serving as the basic units. These pixels are square-shaped and arranged to form the fundamental components of raster data, with each pixel assigned a brightness or color value. Raster data structures are simple, and data can be easily acquired using scanners, satellite imagery, or digital cameras. However, when zoomed in, the lines and edges of objects appear jagged, and increasing the resolution significantly increases the amount of data. Additionally, it is challenging to select or edit objects such as points, lines, and surfaces individually, and raster data cannot handle topological relationships, making spatial relationship analysis impossible. Image data is a typical example of raster data.

Vector data represents the geographic location and shape of objects through coordinates, size, and direction. The vector data model utilizes points, lines, and polygons as basic elements to represent shapes. A point, expressed as a pair of coordinates, indicates a location in space and is a zero-dimensional spatial object. A line, formed by connecting infinite points, represents its position and shape in space as a one-dimensional spatial object with length. A polygon is a two-dimensional geometric element with a closed loop of lines. Vector data allows for smooth and accurate depiction of shapes even when zoomed in or out on a computer, and it compresses data more efficiently than raster models, avoiding the storage of unnecessary details. Additionally, individual objects can be selected and edited. However, the data structure

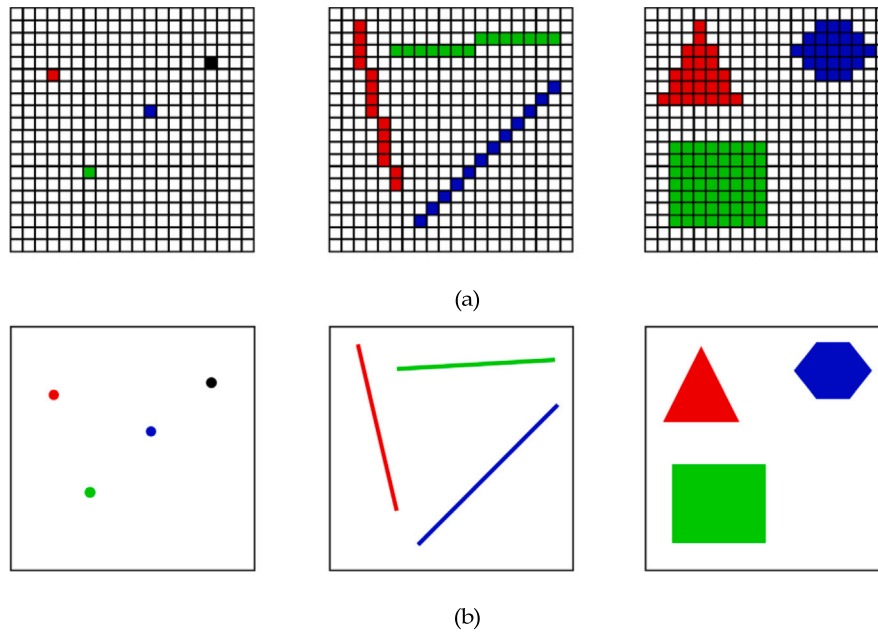


Fig. 1. Representation of points, lines, and polygons in (a) raster and (b) vector format.

is complex, and a vectorization process is required to convert raster data obtained from devices, such as cameras into vector data.

Fig. 1 illustrates the difference in representation methods between raster data and vector data. To represent an object in raster format, each pixel must be utilized. Pixels are arranged in a grid, which makes it challenging to represent the boundaries and edges of objects. As shown in Fig. 1(a), when representing an object in raster format, aliasing occurs, causing the edges to appear jagged, especially when representing lines. For polygons that fill wide areas with pixels, the pixel boundaries are relatively less noticeable compared to lines. However, lines are narrow and long, making pixel discontinuity more pronounced. In pedestrian suspension bridge images, the cables may appear as only a few pixels thick or even less than one pixel, making it difficult to identify the cables in the image. Therefore, this study uses raster data to represent relatively large structures, such as the deck and the pylon, which are represented as polygons in the image, while using the vector format to model cables represented as lines in the image. Detailed explanations of semantic segmentation for the semantic information division of raster data are provided in Section 2.2, and detailed explanations of wireframe parsing for vectorizing structural lines and joints in image data are provided in Section 2.3.

2.2. Semantic segmentation

Semantic segmentation is one of the crucial tasks in the field of computer vision, involving the prediction of semantic information within an image and classifying each pixel accordingly. Traditional image segmentation algorithms were generally performed based on clustering that included additional information from contours and edges [26]. However, the use of deep learning in computer vision, including semantic segmentation, became common after AlexNet [27], a CNN-based architecture for processing image data, won the ILSVRC (ImageNet Large Scale Visual Recognition Challenge) 2012, outperforming other models that utilized machine learning methods. Subsequently, CNN-based models [28–30] demonstrated excellent performance in semantic segmentation tasks. Recently, transformer-based models have been developed. The transformer architecture was initially applied in natural language processing (NLP) and utilizes an attention mechanism [31]. Starting with the ViT [32] model, which integrated transformers into computer vision, transformer-based models [33,34] have shown superior performance in semantic segmentation tasks, surpassing traditional

CNN-based models. However, for transformer-based models to achieve high performance, extensive datasets are required, and on smaller datasets, CNN-based models tend to exhibit higher accuracy. For this reason, this study employs the CNN-based DeepLabv3+ [35].

DeepLabv3+ employs an encoder–decoder architecture for semantic segmentation, specifically applying the Atrous Spatial Pyramid Pooling (ASPP) module in the encoder part. The ASPP module uses parallel atrous convolutions that pool features at different resolutions to capture both the local and surrounding features of each pixel (Fig. 2). Atrous convolution is a filter that performs the convolution operation with spaces between the kernel elements according to the atrous rate, thereby enlarging the receptive field. The high-resolution feature map obtained from the encoder is combined with the low-resolution feature map through the decoder module, which gradually restores spatial information, and is transformed into a segmentation map.

2.3. Wireframe parsing

Wireframe parsing is a field of computer vision that extracts line segments and junctions in an image to vectorize and represent the fundamental boundary structure of an object. Traditional approaches for analyzing geometric information within an image relied on detecting local image features such as corners, edges, SIFT features, and patches. However, these methods had limitations when applied to artificial structures and environments with repetitive patterns or many line segments. The advancement of deep learning technology, particularly the discovery of CNN that brought significant changes to the field of computer vision, contributed to solving these issues. Kun Huang et al. proposed a model that uses CNN to infer wireframes, which are sets of line segments and junctions in an image, and constructed a dataset for training this model [36]. Since then, various models for wireframe parsing have been developed [37–40]. In this study, the HAWPv2 model [41] was used to predict the cables of pedestrian suspension bridges.

Attraction region refers to an area where all pixels within the region are represented by its attraction/projection vector between the pixel and its projection point on the geometry line segment. HAWPv2 utilizes the Holistic Attraction (HAT) field, which enables efficient learning by representing line segments through the distance and angle between points within the attraction region and the line segment. As illustrated

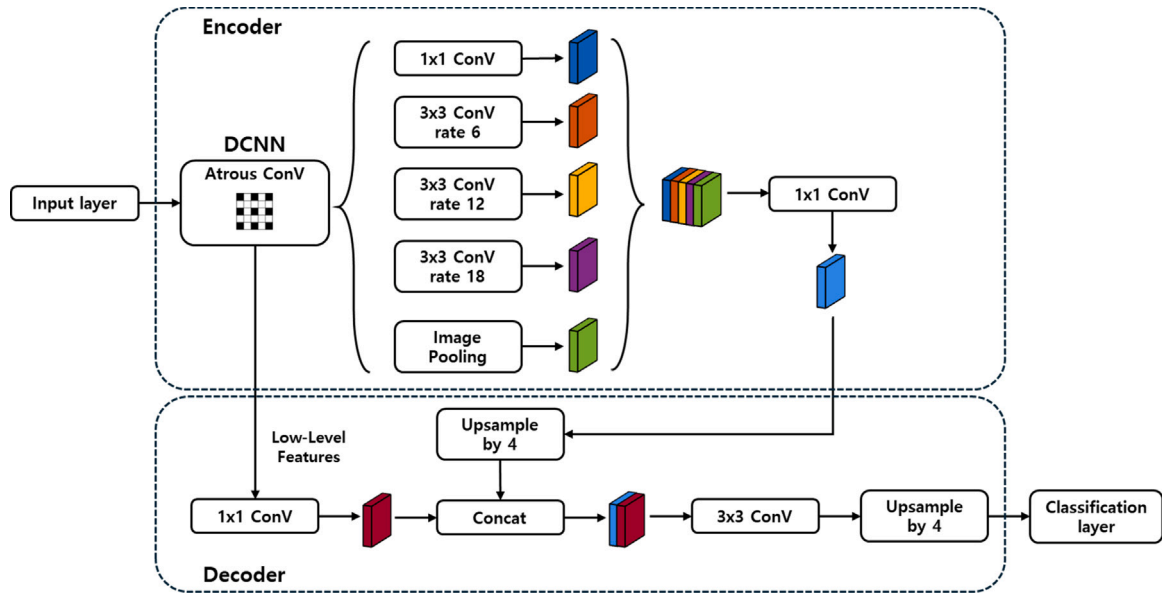


Fig. 2. Architecture of DeepLabv3+.

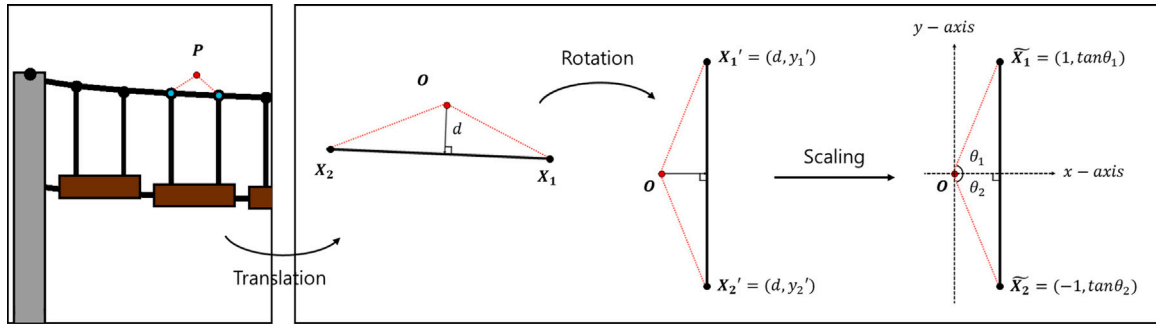


Fig. 3. Illustration of constructing the proposed HAT field representation for wireframes.

in Fig. 3, the HAT field representation for wireframe parsing undergoes three processes: (1) Transformation, (2) Rotation, and (3) Scaling.

Let p be a point within the attraction region and $L = (x_1, x_2)$ be the line segment. First, the point p is translated so that it becomes the origin o of the new coordinate system. Then, the line segment L is rotated such that one endpoint x_1 is positioned in the first quadrant and the line segment L is perpendicular to the x -axis. The rotation angle is denoted as θ , and the angles formed between the x -axis and the two endpoints x_1 and x_2 of the line segment L are denoted as θ_1 and θ_2 , respectively. Finally, the coordinate system is normalized by adjusting the scale so that the distance d between the line segment L and the point p is 1. This normalized representation reduces computational complexity during learning, promoting efficient training. Through this process, the point p within the attraction region of the line segment L can be parameterized as follows:

$$p(L) = (d, \theta, \theta_1, \theta_2)$$

2.4. SfM

Structure from Motion (SfM) is the process of recovering 3D structures and camera poses from a series of images of a structure taken from different viewpoints [42]. SfM has evolved through three approaches: Incremental, Hierarchical, and Global. In this study, Incremental SfM, a technique that processes images sequentially, was applied. The first step of Incremental SfM is the correspondence search, which finds scene overlap in the input images and identifies projections of the same points

in overlapping images. For feature points to be uniquely recognizable across multiple images, they must be invariant to radiometric and geometric changes. To extract such feature points, feature descriptors like SIFT [43] are used. Next, images viewing the same scene part are matched based on the extracted feature points. However, matched images based solely on appearance do not guarantee that the features are mapped to the same scene points. Therefore, geometric verification is performed by estimating the transformation that maps feature points between images from the perspective of epipolar geometry [44]. In epipolar geometry, the relationship with the moving camera can be expressed through the essential matrix E (calibrated) or the fundamental matrix F (uncalibrated). Correspondences from matching are often contaminated with outliers, so outlier removal algorithms like RANSAC [45] are used. The second step is Incremental Reconstruction. The input to the reconstruction is a scene graph, and the outputs are camera pose estimates for registered images and the reconstructed scene structure as a set of points. To initialize the model, a two-view reconstruction is performed by selecting the initial two images to be registered [46]. It is important to select images with significant overlap from multiple cameras to ensure an accurate and robust reconstruction. Next, the Perspective-n-Point (PnP) problem [45] is solved through matching the feature points of the registered images to estimate the camera poses and register new images to the current model. Subsequently, the 3D points of the newly registered images are added to the previously registered 3D points through triangulation. As the previous processes include errors, bundle adjustment is performed to minimize the reprojection error, thereby adjusting the estimated camera poses

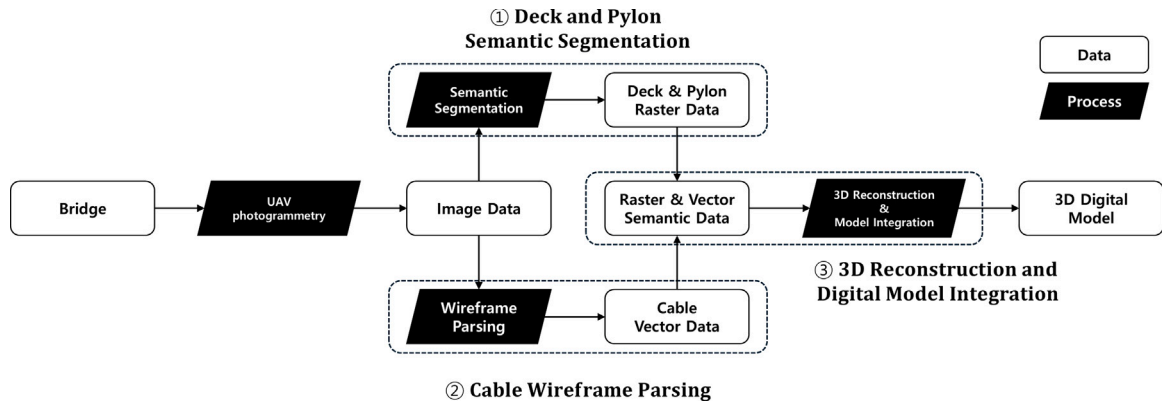


Fig. 4. Flowchart of proposed Hybrid SSfM.

and 3D points. The restored camera pose T can be represented by the following equation.

$$T = [R|t]$$

Here, R is the 3 by 3 rotation matrix, and t is the 1 by 3 translation vector. These are used to project 3D points from the world frame to the image. A 3D point expressed in the homogeneous coordinate system as $P = [X \ Y \ Z \ 1]^T$ and a corresponding 2D point on the projected image plane as $p = [x \ y \ 1]^T$ can be related as follows:

$$s \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = K[R|t] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Here, an arbitrary s is a scale factor, and K , referred to as the intrinsic parameter.

3. System development

Fig. 4 shows an overview of the proposed Hybrid SSfM for the digital transformation of pedestrian suspension bridges. The proposed system utilizes image data of pedestrian suspension bridges obtained through UAVs aerial photogrammetry. The basic pipeline consists of three phases. As shown in Fig. 4, Phase 1 is Deck and Pylon Semantic Segmentation (Sec 3.1). In this phase, relatively large components, such as the deck and pylon, are represented in raster format through semantic segmentation. Phase 2 is Cable Wireframe Parsing (Section 3.2). In this phase, cables, which are difficult to represent in the raster format, are represented in the vector format through wireframe parsing. Finally, Phase 3 is 3D Reconstruction and Model Integration 3.3. In this phase, the structural semantic information obtained in raster and vector formats in the previous phases is converted into 3D and the models are integrated into a unified 3D model.

3.1. Phase 1: Deck and pylon semantic segmentation

The objective of Phase 1 is to represent the semantic information of relatively large structural components, specifically the deck and pylon, within a pedestrian suspension bridge in raster format through semantic segmentation. This is achieved by modifying the classification layer of a pre-existing semantic segmentation model through transfer learning. The model uses images captured by UAVs of the pedestrian suspension bridge to segment the Deck and Pylon. Transfer learning involves tuning a pre-trained network to perform tasks using new but less data. By employing this method, it is possible to leverage the robust feature extraction capabilities of a model trained on a large dataset, allowing for high performance with less data and reduced training time. To effectively implement transfer learning, the pre-trained model must be adjusted to align with the specific dataset at hand. In semantic

segmentation tasks, the output tensor size of the classification layer is $H \times W \times C$, where H is the height of the image, W is the width, and C is the number of classes (Fig. 5). This output tensor corresponds to each pixel in the image, assigning semantic segmentation scores for each class at every position. The class with the highest score at each pixel is determined as the final prediction result. To facilitate pixel-wise prediction through semantic segmentation, the width and height of the images in the dataset are adjusted accordingly. Additionally, the classes are defined as pylon, deck, and background, with the depth of the classification layer adjusted to the number of classes, allowing each pixel to be classified into one of the three categories, and visualizing the class that scores the highest.

In the application of transfer learning to deep neural networks, the strategy of freezing or selectively fine-tuning specific layers is employed to optimize the model for new datasets. Freezing layers preserves the pre-trained weights, effectively retaining the neural network's previously learned feature extraction capabilities. Conversely, fine-tuning involves adjusting the weights of specific layers to enable the model to adapt to the characteristics of the new dataset. When approaching a large-scale dataset that significantly differs in characteristics from the dataset used for initial training, updating the weights across all layers becomes necessary to accommodate the distinctive features of the new data. However, when the new dataset is smaller and resembles the previously trained dataset, freezing the intermediate layers while only fine-tuning the classifier layers is generally advantageous. This approach helps maintain the robust feature extraction capabilities acquired from the larger dataset, enabling the model to efficiently perform the desired tasks. For training a model to segment the pylon and deck from pedestrian suspension bridge images, image data is categorized into three distinct classes: pylon, deck, and background, as illustrated in Fig. 6. Deck and Pylon constitute only a small portion of the image compared to the Background, resulting in class imbalance. To address this issue, methods such as the oversampling of minority classes, the undersampling of majority classes, class weight adjustment, and the Synthetic Minority Over-sampling Technique (SMOTE) can be utilized. Among these, this study employs class weight adjustment by using reciprocal of the pixel counts per class as weights to balance each class, thereby resolving the class imbalance issue.

To enhance the performance of semantic segmentation models via transfer learning, it is critical to select appropriate hyperparameters such as learning rate, batch size, epochs, and optimizer. The learning rate is a coefficient that dictates the magnitude of weight updates during model training, thereby controlling the rate at which learning progresses. An improperly sized learning rate can lead the model to converge to local minima or diverge, thereby failing to converge to the global minimum. The batch size represents the number of data samples included in a small subgroup when the entire training dataset is divided into multiple batches. The size of the batch size influences the speed and stability of learning and is impacted by computational

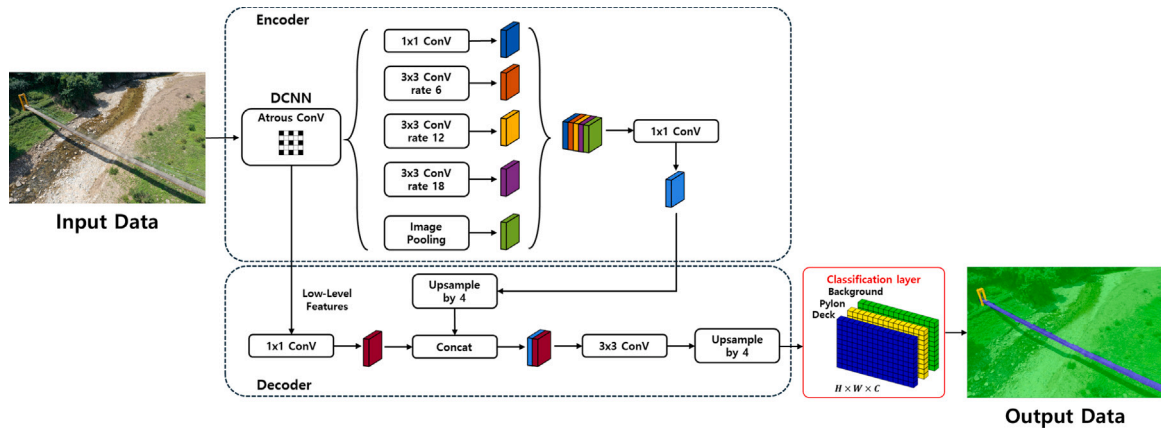


Fig. 5. Architecture of the proposed deck and pylon semantic segmentation network.

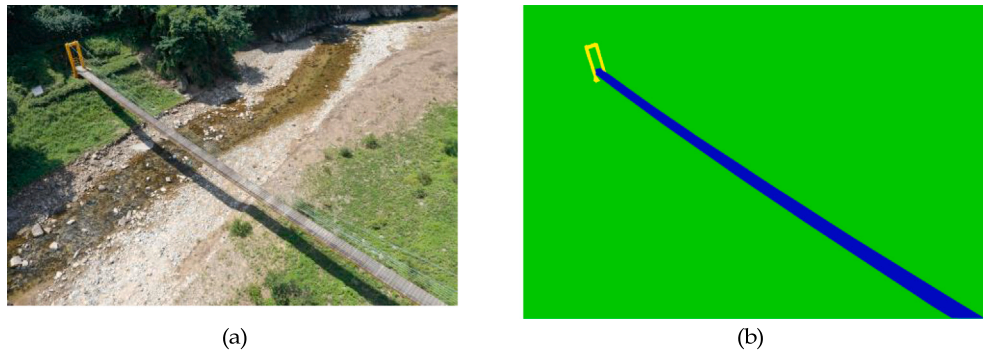


Fig. 6. (a) Image data. (b) Label data.

resources. Epochs denote the number of times the complete dataset is used for training. Insufficient epochs can result in underfitting, whereas too many epochs can lead to overfitting. The optimizer is an algorithm that adjusts the model's parameters to minimize the loss function value for the given data. An example of an optimizer is the Stochastic Gradient Descent with Momentum (SGDM), which trains the model using randomly selected mini-batches. It incorporates momentum by considering both the direction and magnitude of previous gradient updates in the current update. This approach helps to stabilize the training process and promotes smoother convergence.

3.2. Phase 2: Cable wireframe parsing

Phase 2 of the proposed system focuses on Wireframe Parsing for cable modeling. This technique enables the representation of cables in vector format, which is challenging to represent using the raster format. To predict the coordinates of cables within images, the proposed system employed a wireframe parsing model adapted through transfer learning. This process utilizes the dimensions of the image file (height and width), the coordinates of the endpoints of line segments representing the main cables, and the connectivity information between each junction and its linked junctions. Fig. 7 illustrates the architecture of the cable wireframe parsing network. When wireframe parsing is applied to an image, it predicts a wireframe that includes both junctions and line segments, and outputs their corresponding confidence scores. During the wireframe parsing process, the confidence score indicates the detection accuracy of the line segments and junctions within the image. However, the wireframe parsing might produce some false-positive line segments and/or junctions.

The two common false-positive results are shown in Fig. 8(b); case 1: three or more line segments are connected at one junction, and case 2: a single junction is identified as multiple junctions. For most

pedestrian suspension bridges, the main cable is a single continuous cable. Therefore, when the main cable is represented with wireframes (i.e. multiple junctions and line segments), each junction point of the cable should be connected to one line segment for the first and last junctions, and two line segments for all junctions besides the first and last junctions. However, as illustrated in case 1, some junctions are connected to more than three line segments due to false positives. To remove the false positives, the proposed method retains only the top two line segments with the highest confidence scores, and removes the other segments with low confidence scores.

Let J_i be i th junction of the main cable, and L_{mn} be the line segment of the cable connecting the junction J_m with J_n as shown in Fig. 9. When three or more-line segments are connected at J_i , then the proposed method compares C_{mn} , the confidence scores of the line segments L_{mn} where either $m = i$ or $n = i$, and removes the line segments besides the line segments with top 2 confidence scores.

Another example of a false-positive occurs when a single junction is mistakenly identified as multiple junctions (Fig. 10). Wireframe parsing often results in such misidentification, particularly when the junction is not clearly distinguishable from the background. However, simply applying a threshold to the distances between adjacent junctions does not fully resolve this issue, as 2D images are subject to perspective distortion. Junctions closer to the camera appear farther apart, while those farther away seem closer together, making it challenging to apply a fixed threshold for accurate detection.

To address this issue, a method based on the ratio of distances between junctions was utilized. Let J_i denote the given junction, J_p the closest junction to J_i , and d_{ip} the length of the line segment L_{ip} . Similarly, let J_q represent the second closest junction to J_i , J_r the closest junction to J_q , and d_{qr} the length of the line segment L_{qr} . The ratio β_i , defined by the two distances d_{ip} and d_{qr} , is expressed by the

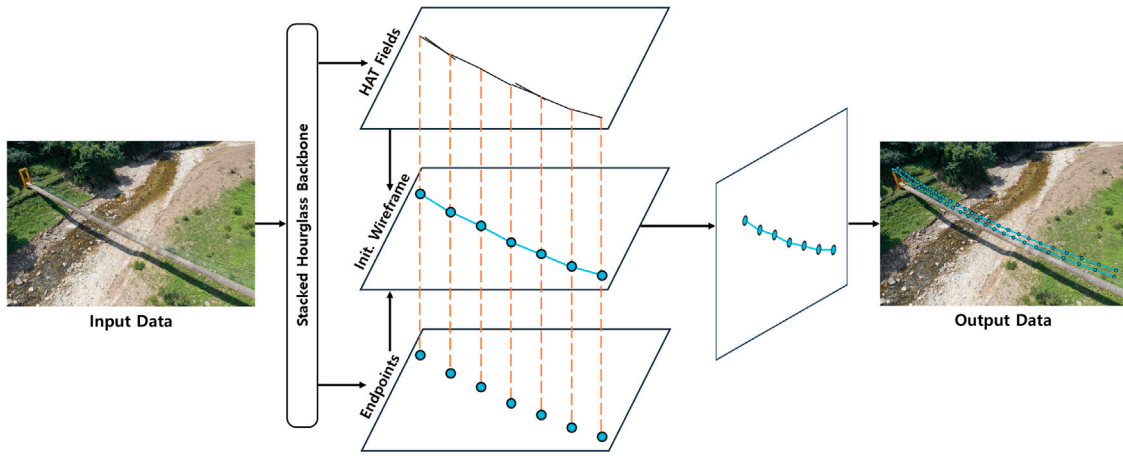


Fig. 7. Architecture of the proposed cable wireframe parsing network.

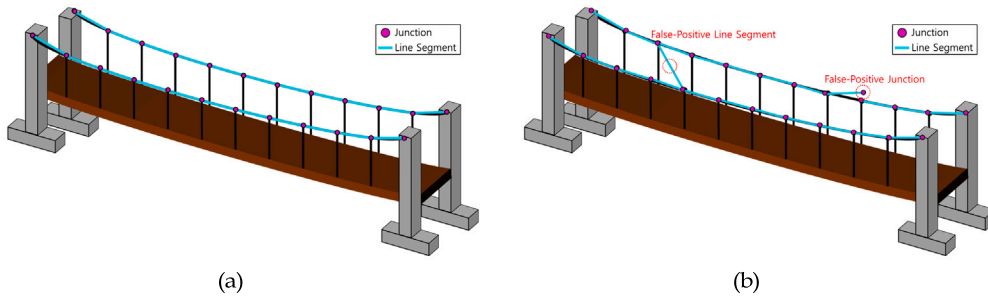


Fig. 8. (a) Cable labeling and (b) Cable wireframe parsing.

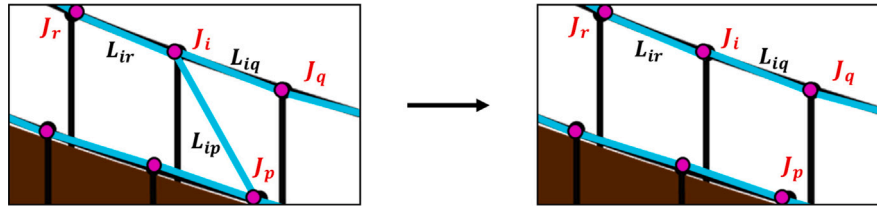


Fig. 9. Example of false positive case 1: three or more line segments are connected at one junction.

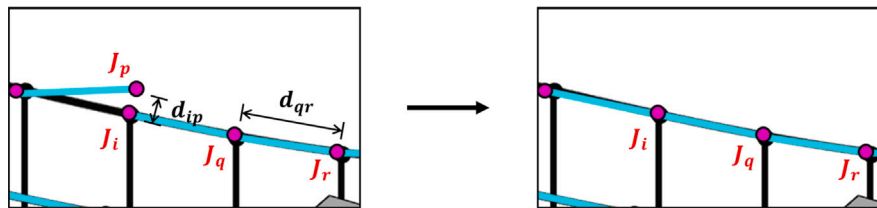


Fig. 10. Example of false positive case 2: a single junction is identified as multiple junctions.

following equation:

$$\beta_i = d_{qr} / d_{ip} \quad (1)$$

Fig. 11(a) shows the β values of all junctions sorted in descending order. Group 1 demonstrates that β is close to 1, indicating no false positives, as the distances between adjacent junctions are similarly distorted due to perspective effects. Group 2 consists of junctions where a single junction is misclassified as two separate ones, and because the value of d_{ip} is small, β is calculated to be greater than 1. Group 3 comprises junctions located near the false-positive junctions in Group 2,

where the small d_{qr} results in a lower β value. In this study, the Theil–Sen estimator algorithm was applied to identify false-positive junctions in Group 2.

As shown in Fig. 11(b), the Theil–Sen estimator is robust against outliers and generates a regression line through the following steps: (1) calculating all possible slopes from the data, (2) determining the median of the calculated slopes, (3) using the median slope to compute the y-intercept, and (4) generating the regression line based on the slope and intercept. Given two data points (x_u, y_u) and (x_v, y_v) , the slope

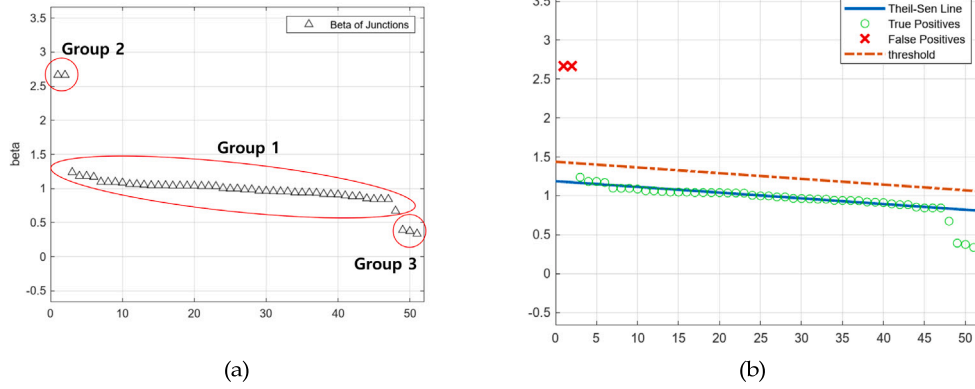


Fig. 11. (a) β of junctions in descending order and (b) Theil-Sen Estimator for Identification of False Positives.

$m_{u,v}$ and y-intercept $b_{u,v}$ are derived using the following equations:

$$m_{u,v} = \frac{y_v - y_u}{x_v - x_u} \quad (2)$$

$$b_{u,v} = y_u - m_{u,v} \cdot x_u \quad (3)$$

The calculated Theil-Sen line can be used to identify false positives that are outside the threshold range. The detected false positives refer to instances where a single junction has been mistakenly identified as two separate junctions. If junctions J_i and J_p are classified as false-positive junctions by the Theil-Sen estimator, their confidence scores C_i and C_p are compared to decide which junction to eliminate. The junction with the lower confidence score is then removed. Finally, the line segments connected to the removed junction are reconnected to the remaining junction.

3.3. Phase 3: 3D reconstruction and digital model integration

The objective of Phase 3 is to transform the component information of the pedestrian suspension bridge obtained in previous phases into a 3D format and to integrate the two models. Initially, SSfM technology is employed to transform the semantic information represented in raster format, acquired from Phase 1 from 2D to 3D. SSfM is an approach for assigning semantic information to a point cloud generated through SfM. The method to identify the 2D point $p_i = [x_i \ y_i]$ in the reference images, in the i th image, which is used as a reference to predict the 3D point $P = [X \ Y \ Z]$, is as follows.

$$s \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = K[R_i|t_i] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad (4)$$

Where, s is an arbitrary scale factor, K denotes the intrinsic parameters, and $[R_i|t_i]$ denote the extrinsic parameters of the i th image. Knowing the position of a 2D point referenced by a 3D point enables classification of the 2D point using semantic segmentation. However, a single 3D point may correspond to pixels across multiple images, and there is no guarantee that the classes of these pixels, obtained through semantic segmentation, will be consistent. This uncertainty in class assignment appears as noise within the segmented points cloud. To address this, the classes of all 2D points corresponding to a single 3D point are predicted, and the most frequently occurring class is selected as the representative class for that 3D point. Furthermore, this study established a 75% threshold to enhance the precision of point cloud segmentation. Points were classified as background and noise was removed if the most prevalent class constituted less than this threshold percentage among all predicted classes.

However, the 3D model obtained through SSfM does not include the cables, which are vital components of the pedestrian suspension bridge.

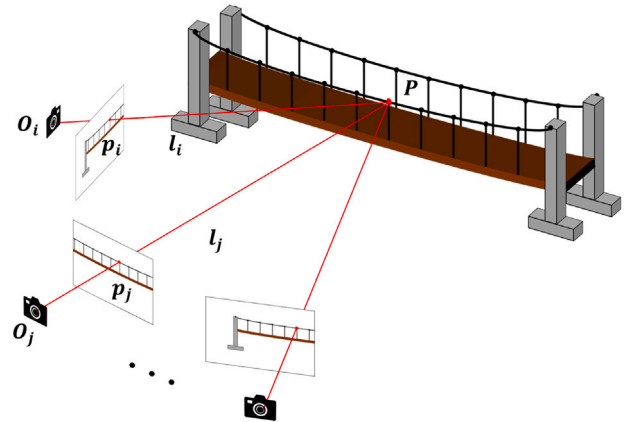


Fig. 12. Multiview triangulation.

To reconstruct the 3D structure from the 2D information of cables obtained in Phase 2, the following three processes are implemented: (1) camera pose estimation, (2) 3D junction points reconstruction using triangulation, and (3) outlier removal. In the first step, the extrinsic parameters obtained from the SfM process are utilized for camera pose estimation. These extrinsic parameters are employed to project 3D points from the world frame to the camera frame. Conversely, the extrinsic parameters for transforming from the camera frame back to the world frame can be determined through the following mathematical formula.

$$R_{w,i} = R_i^T \quad (5)$$

$$t_{w,i} = -R_i^T \cdot t_i \quad (6)$$

Here, $R_{w,i}$ and $t_{w,i}$ are the rotation matrix and translation vector, respectively, used to transform 3D points from the camera frame to the world frame for the i th image. For multiview triangulation, the current pose and relative pose of the camera are required. Let T_i denote the camera pose of the i th image and T_j denote the camera pose of the j th image. The current pose and relative pose are defined by the following equations:

$$\text{current pose} = T_i = \begin{bmatrix} R_{w,i} & t_{w,i} \\ 0_{3 \times 1} & 1 \end{bmatrix} \quad (7)$$

$$\text{relative pose} = T_j \times T_i^{-1} \quad (8)$$

During the process of 3D junction point reconstruction using triangulation, the estimated camera poses, along with the 2D coordinates of the cables obtained from Phase 2, are utilized. Triangulation is used to

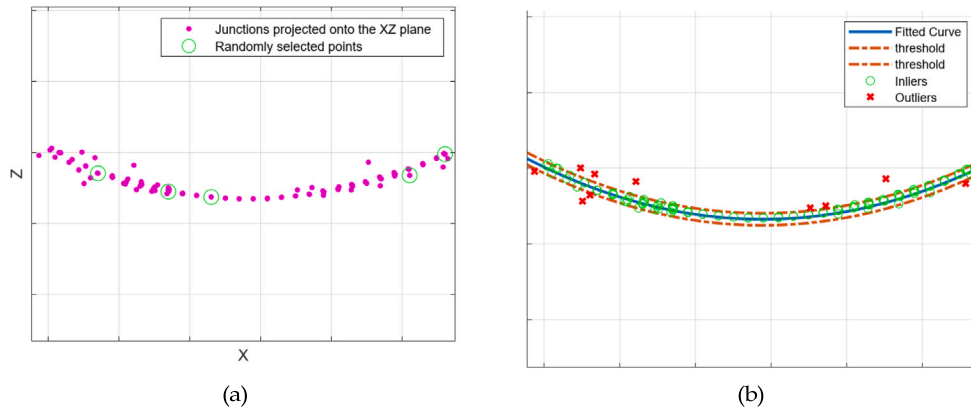


Fig. 13. Stages of the RANSAC Algorithm: (a) Random sampling, (b) Model fitting and outlier removal.

determine the position of a 3D point by using its projections in two or more images. Given the camera poses for the i th and j th images, the lines l_i and l_j can be calculated, which connect the origins O_i and O_j of each camera to the 2D coordinates p_i and p_j of the cables, as depicted in Fig. 12. The 3D point P can be determined as the intersection of lines l_i and l_j . Subsequently, the accuracy is refined using bundle adjustment, which optimizes the 3D point by minimizing the reprojection error.

Through this process, the 3D spatial coordinates of the cable junctions can be estimated. However, the estimated 3D coordinates of the cables include noise. Due to the need for precise prediction of each junction in cable modeling, several post-processing steps must be implemented to address this issue. Initially, the pedestrian suspension bridge undergoes a 3D rotational transformation. First, the endpoints of the created cable model are manually selected and then moved to the origin. Next, the angles between the cable's longitudinal axis and each axis are calculated. Afterward, a rotation transformation matrix is applied to the point cloud. The rotation angles around the x -axis, y -axis, and z -axis are denoted as ϕ , θ , and ψ , respectively, and the formula for the three-dimensional rotation matrix is as follows.

$$R_{xyz} = R_x(\phi)R_y(\theta)R_z(\psi) \quad (9)$$

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix} \quad (10)$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \quad (11)$$

$$R_z(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (12)$$

After applying a 3D rotation transformation to the cable junctions, noise is removed. To reduce noise, the RANSAC (RANdom Sample Consensus) algorithm is employed. RANSAC is an algorithm used to estimate a model while minimizing the influence of outliers. First, a model is estimated by randomly selecting sample data from the dataset. In this study, to estimate a model suitable for the cable, the 3D points of the cable were projected onto the xz -plane, and the coefficients of a quadratic polynomial were estimated using the Least squares approximation based on the selected data. Then, data points within a predefined distance threshold from the model are classified as inliers, and their count is recorded. This process is repeated to select the model that maximizes the number of inliers and removes outliers. Fig. 13 illustrates the process of the RANSAC algorithm.

Subsequently, clustering is performed to adjust the number of junctions using the K-means algorithm. With the original data provided (Fig. 14(a)), the initial step in K-means clustering involves determining the number of groups into which the data will be clustered and setting

Table 1
Specification of Mavic Pro.

Specification	Value
Size (when folded)	H83 × W83 × L198 (mm)
Hovering accuracy	±0.1 m (Vertical), ±0.5 m (Horizontal)
Weight	743 g
Lens	f/2.2
Max speed	65 km/h
Image sensor	1/2.3" CMOS
Max flight Time	Approx 27 min
Focal length	26 mm
Max service ceiling above sea level	5000 m
Image resolution	4000 × 3000
FOV	78.8°

the initial centroids (Fig. 14(b)). In the following step, all data points are assigned to the nearest centroid based on distance. Once the assignments are completed, the mean point of the data within each cluster is recalculated to reset the centroid of the cluster (Fig. 14(c)). This process is repeated to reassign the data to new centroids and reset the centroids again. When there is no further movement of the centroids, as shown in Fig. 14(d), the data within each cluster are consolidated at the cluster centroids. The k-means clustering algorithm is sensitive to the threshold value. Applying wireframe parsing allows for the extraction of the number of junctions within an image, which was subsequently utilized as the threshold value for k-means clustering. This procedure allows for the acquisition of adjusted 3D coordinates of the cable by managing the number of junctions.

The final process in Phase 3 is model integration. The previous steps describe methods for transforming 2D information of the deck and pylon, obtained through semantic segmentation in raster format, and the 2D information of cables, obtained through wireframe parsing in vector format, into 3D reconstructions. Each component model generated through SSfM and multiview triangulation techniques is integrated to produce a 3D model of the pedestrian suspension bridge, which comprises the deck, pylon, and cables.

4. Validation

To validate the proposed Hybrid SSfM, a field experiment was conducted on the Yangban Suspension Bridge located in Goesan, Chungcheongbuk-do, South Korea. Image data of the pedestrian suspension bridge is required for the proposed system, and to obtain this data, UAVs aerial photogrammetry was conducted. The equipment used for aerial photogrammetry was the DJI Mavic Pro (Fig. 15). The specifications of the equipment are detailed in Table 1. A total of 80 images of the pedestrian suspension bridge were obtained through aerial photogrammetry.

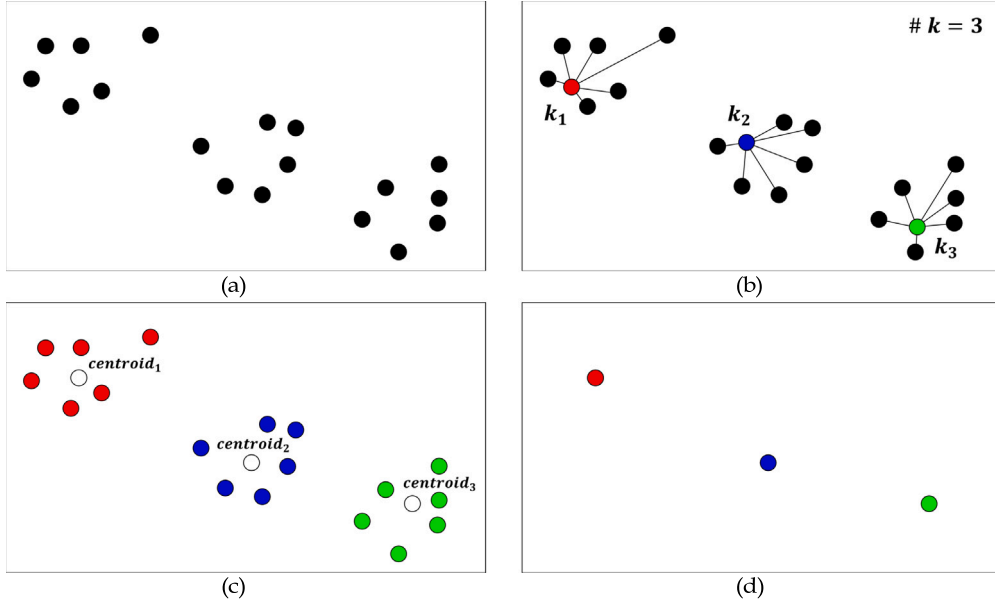


Fig. 14. Stages of the k-means Clustering Algorithm. (a) Original data, (b) Randomly centroid choosing and clustering, (c) New centroid calculation and clustering, and (d) Consolidation of clustered data when centroids cease to move.



Fig. 15. DJI Mavic Pro.

In this study, DeepLabv3+ with a pre-trained ResNet-18 backbone was used for semantic segmentation of pedestrian suspension bridge images. For transfer learning, a total of 50 images of the pedestrian suspension bridge were used, of which 80% were allocated for training and 20% for validation. As described in 3.1, appropriate hyperparameters are crucial for effective transfer learning. To ensure rapid and stable weight updates, the SGDM optimizer was employed. To prevent overfitting, the maximum number of epochs was limited to 30. The initial learning rate was set to 0.001, and it was decreased by a factor of 0.3 every 10 epochs to converge to the optimal value. Additionally, the mini-batch size was set to 2.

To evaluate the results of semantic segmentation, a confusion matrix is utilized. Utilizing a confusion matrix allows for the calculation of evaluation metrics such as Accuracy and Intersection over Union (IoU). Let True Positive, False Negative, and False Positive be represented as TP, FN, and FP, respectively. Accuracy and IoU can be expressed using the following equations.

$$Accuracy = \frac{TP}{TP + FN} \quad (13)$$

$$IoU = \frac{TP}{TP + FN + FP} \quad (14)$$

Previous studies on railroad bridges [23] segmented all structural components using only semantic segmentation. In contrast, the proposed system separately employs the wireframe parsing algorithm specifically for accurate cable modeling. To compare the effects of these methods, case 1, employing the conventional SSfM approach,

was segmented into five classes: deck, pylon, cable, junction, and background, compared to the proposed Hybrid SSfM, case 2, which segmented into three classes: deck, pylon, and background. The results of the model training for case 1 and case 2 are shown in Table 2, and their confusion matrices are as illustrated in Fig. 16. The values on the right of the confusion matrices represent the precision for each class, and the values at the bottom indicate the recall for each class.

As shown in Table 2 and Fig. 16, the IoU and Recall values for the Cable and Junction classes in case 1 are notably low. This is attributed to the challenges of accurately predicting cables and junctions in pedestrian suspension bridge images, where their width can be as small as a few pixels or even less than a single pixel. Unlike Accuracy, which considers only the proportion of correctly predicted pixels relative to the total pixel count, IoU assesses the overlap between the predicted and actual areas. Consequently, the smaller regions occupied by the Cable and Junction resulted in lower IoU values. Fig. 17 illustrates the label data used for model training in case 1 and the predictions made by the model. As shown in Fig. 17, predicting cables and junctions through semantic segmentation proves to be challenging. To address these issues, the proposed system employs wireframe parsing to segment cables into vector format, rather than relying on raster-based segmentation through semantic segmentation.

To predict the locations of cables within the images, the wireframe parsing model, HAWPv2, was employed. A total of 80 pedestrian suspension bridge images were used for the transfer learning of the model, with 75 images allocated for training and 5 for validation. The hyperparameters for transfer learning were set as follows: the learning rate was set to 0.0004, the maximum number of epochs was set to 30, and the optimizer used was ADAM. Additionally, the maximum numbers of predicted junctions and line segments were set to 250 and 2500, respectively.

The performance of the wireframe parsing model used to predict the position of cables in images is evaluated using structural average precision (sAP) [37,41]. sAP is defined to be the area under the precision recall curve computed from a scored list of the detected line segments on all test images. A detected line segment is considered a TP if it satisfies the following equation.

$$\min_{(i,j)} \|x_1 - x_i\|^2 + \|x_2 - x_j\|^2 \leq \vartheta_L \quad (15)$$

Where, $(i, j) \in (1, 2), (2, 1)$, ϑ_L is a predefined threshold, $L_{gt} = (x_1, x_2)$ are the ground-truth line segments, and $L = (x_i, x_j)$ are the detected

Table 2
Semantic segmentation results.

Class	5-class semantic segmentation (for conventional SSfM)		3-class semantic segmentation (for Hybrid SSfM)	
	Accuracy (%)	IoU (%)	Accuracy (%)	IoU (%)
Deck	96.70	80.42	99.32 (+2.62)	87.32 (+6.90)
Pylon	96.34	52.38	98.17 (+1.83)	60.36 (+7.98)
Background	98.07	98.06	99.44 (+1.37)	99.42 (+1.36)
Cable	86.05	26.06	–	–
Junction	74.55	4.17	–	–

5-class semantic segmentation (for conventional SSfM)

True Class \ Predicted Class	Background	Cable	Deck	Junction	Pylon
Background	39262365	346828	280244	108451	36985
Cable	415	134111	2309	17162	1849
Deck	3037	9750	1352881	29676	3707
Junction	12	2023	69	6890	248
Pylon	450	210	613	595	49120

100.0%	27.2%	82.7%	4.2%	53.4%
0.0%	72.8%	17.3%	95.8%	46.6%

(a)

3-class semantic segmentation (for Hybrid SSfM)

True Class \ Predicted Class	Background	Deck	Pylon
Background	39948352	194900	31274
Deck	7272	1412044	2398
Pylon	447	537	52776

100.0%	87.8%	61.0%
0.0%	12.2%	39.0%

(b)

Fig. 16. Confusion matrices for the semantic segmentation results of (a) case 1 and (b) case 2.

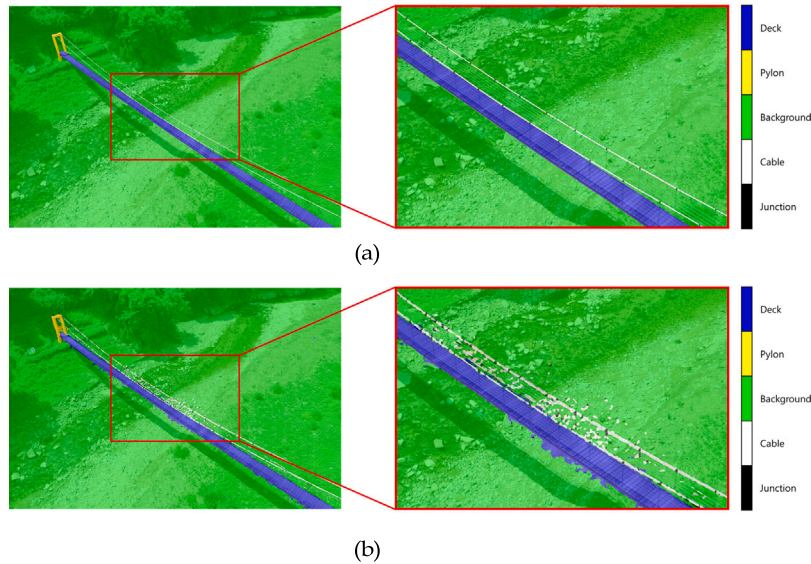


Fig. 17. (a) Ground-truth label data, (b) predicted label data for case 1.

line segments. When the sizes of the detected wireframe and ground-truth wireframe are adjusted to 128×128 , the sAP scores are denoted as sAP_5 , sAP_{10} , and sAP_{15} for θ_L values of 5, 10, and 15, respectively. The sAP values of the trained model are 74.8, 84.5, and 86.3 for sAP_5 , sAP_{10} , and sAP_{15} , respectively.

After applying wireframe parsing to 80 images of pedestrian suspension bridges, we observed 67 instances of false positive line segments and 8 instances of false positive junctions. The two methods for removing false positives mentioned in Phase 2 successfully eliminated all false positives identified in the dataset. The Fig. 18, illustrates the process of removing false positives from pedestrian suspension bridge images following wireframe parsing.

In this study, COLMAP (<https://github.com/colmap/colmap>) was employed to estimate camera poses and generate a point cloud for a pedestrian suspension bridge using SfM. Following this, triangulation for cable modeling was conducted using the 2D coordinates of the cables obtained through wireframe parsing and the camera poses estimated via SfM. To evaluate the performance of the cable's 3D digital

Table 3
Prediction results of sag ratio.

Ground truth	Conventional SSfM		Hybrid SSfM	
	Sag ratio	RE (%)	Sag ratio	RE (%)
0.0579	0.0629	8.62	0.0581	0.42 (−8.2)

model, root mean square error (RMSE), mean absolute error (MAE), and the sag ratio were measured. The RMSE and MAE between the ground-truth junctions in vector format and those obtained through the proposed Hybrid SSfM were 0.2208 m and 0.1847 m, respectively. The term sag refers to the vertical deflection of the cable due to its self-weight, and the sag ratio is defined as the sag divided by the horizontal span between two pylons. The ground-truth sag ratio was 0.0579, while the sag ratio obtained from the conventional SSfM was 0.0629, resulting in a relative error(RE) of 8.6249%. In contrast, the sag ratio and relative error obtained through the proposed Hybrid SSfM were 0.0581 and 0.4196%, respectively, demonstrating a more accurate

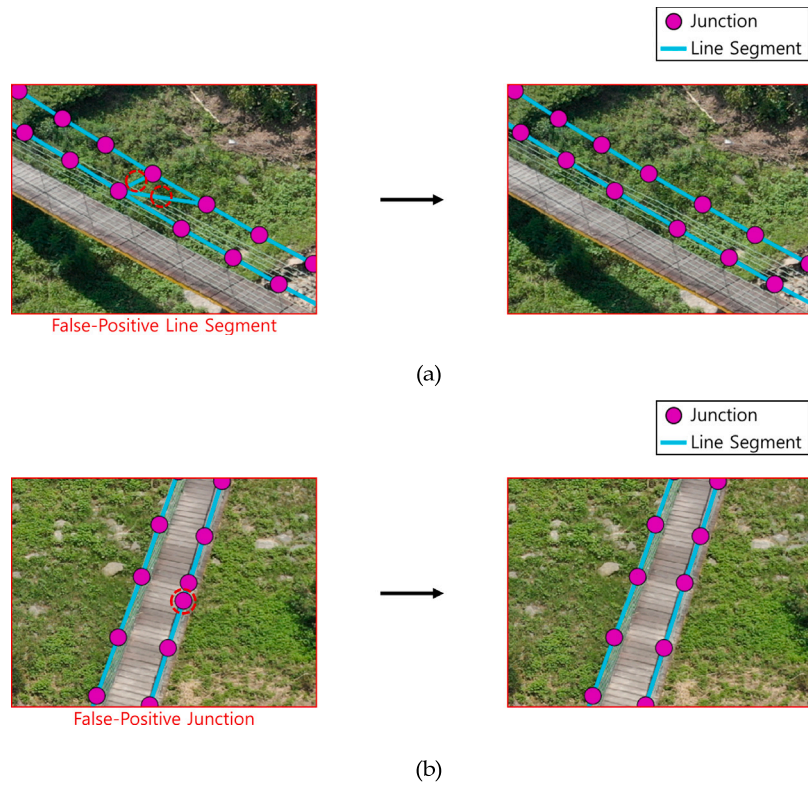


Fig. 18. Removal of (a) false positive line segment and (b) false positive junction.

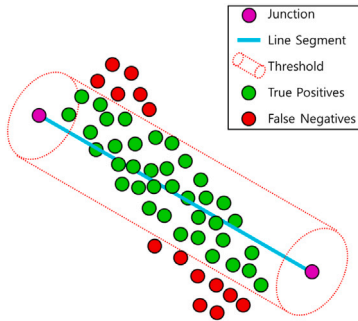


Fig. 19. True positive classification based on distance to line segments.

measurement compared to the conventional SSfM. Table 3. provides a summary of these results, highlighting the improved accuracy of the Hybrid SSfM method.

To validate the results of SSfM obtained through the proposed method, the Accuracy and IoU of point clouds generated by conventional SSfM and Hybrid SSfM were measured. Since the proposed system represents cables and junctions in vector format, the approach of comparing point classes is not applicable for evaluating the method. To address this, a method was employed where ground-truth points within a threshold distance from the predicted line segments, as illustrated in Fig. 19, are classified as TP. For this study, the threshold was set to 0.1 m.

As evidenced in Table 4, it is clear that the conventional SSfM produces cable and junction models with lower accuracy and IoU. In contrast, the cable and junction models obtained through the Hybrid SSfM method demonstrate significantly higher accuracy, indicating an improvement in performance. Fig. 20(a) represents the ground-truth 3D model of the pedestrian suspension bridge, while 20(b) and 20(c) display the 3D models of the bridge generated using traditional SSfM

Table 4
SSfM results.

Class	Conventional SSfM		Hybrid SSfM	
	Accuracy (%)	IoU (%)	Accuracy (%)	IoU (%)
Deck	90.35	88.32	90.53 (+0.18)	87.32 (−1.00)
Pylon	56.30	49.40	61.49 (+5.19)	54.90 (+5.50)
Background	99.92	99.44	99.89 (−0.03)	99.40 (−0.04)
Cable	28.81	28.27	93.29 (+64.48)	–
Junction	2.37	2.36	65.00 (+62.63)	–

and the proposed Hybrid SSfM, respectively. While the model created through conventional SSfM exhibits discontinuities in the cables, the model utilizing Hybrid SSfM successfully represents the cables in a vector format without discontinuities. This demonstrates that the proposed method, which utilizes both semantic segmentation and wireframe parsing, is more suited for the digital transformation of pedestrian suspension bridges compared to methods that only employ semantic segmentation.

5. Discussion

5.1. Limitation

While the proposed approach shows promising results, there are several challenges that remain open for future investigation. The challenges associated with the deck and pylon modeling, wireframe parsing, UAV flight paths, and evaluation metrics for vector-format cable representation will be discussed in the sections below.

(1) Deck and Pylon Modeling. The proposed method showed significant enhancements in the modeling of cables and junctions with improved predictions in SSfM and Sag ratio, compared to conventional methods. However, the point clouds for decks and pylons, generated using the same techniques as the conventional approach, did not achieve significant improvements. Particularly, many background pixels were

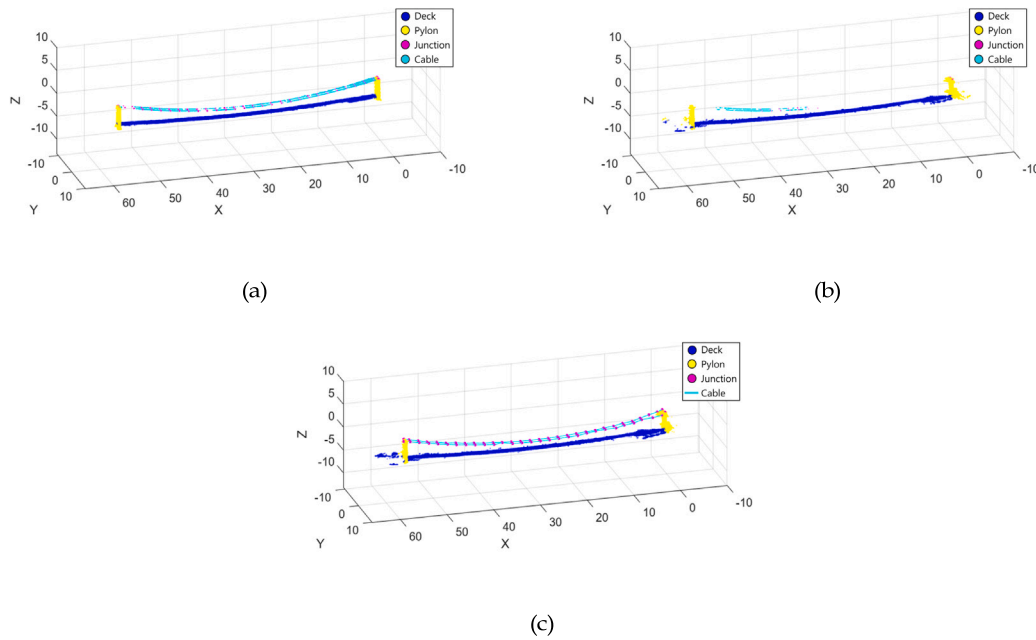


Fig. 20. Comparison of pedestrian suspension bridge 3D models: (a) Ground truth, (b) conventional SSfM, and (c) Hybrid SSfM (Proposed Method).

incorrectly predicted as bridge components in the semantic segmentation process, leading to erroneous predictions for deck and pylon points. The enhancement of their accuracy remains a task for future work. It is anticipated that these challenges can be addressed by utilizing more advanced semantic segmentation models and increasing the training dataset with a larger number of images. These enhancements are anticipated to mitigate the current limitations in the accuracy of deck and pylon modeling in future studies.

(2) Wireframe parsing. The proposed method showed considerable improvement over existing techniques in sag ratio prediction, achieving a relative error of just 0.42%. However, the cable modeling within this method significantly depends on the outcomes of wireframe parsing. This dependence introduces several vulnerabilities, as inaccuracies in wireframe parsing can lead to compounded errors in subsequent stages, including SfM camera pose estimation and the triangulation process. Due to the intrinsic requirement of accurately representing each junction in cable modeling, this research required multiple post-processing steps to mitigate issues with inaccurate wireframe parsing. These included the removal of false positive line segments and junctions, outlier elimination via RANSAC, and adjustment of junction counts using k-means clustering. Looking ahead, it is expected that employing more sophisticated wireframe parsing algorithms and gathering larger datasets will minimize the necessity for such extensive post-processing.

(3) UAV flight paths. In this research, we employed a circular flight path. This circular flight path, commonly used for acquiring UAV-based imagery, may lead to occlusions due to vegetation and other environmental factors, which can obscure parts of the structure and limit point cloud generation in areas not visible from the camera's perspective. As a result, the digital model created might not comprehensively represent the entire pedestrian suspension bridge. It is anticipated that the implementation of optimization algorithms for flight paths and capture angles in future work will enable the creation of more precise digital models of pedestrian suspension bridges.

(4) Evaluation metrics for vector-format cable representation. In this study, we evaluated the vector-format cable representations by measuring the distances between the ground truth cable point clouds and the modeled line segments. Points exceeding a predefined threshold distance were classified as FN, while those below this threshold were classified as TP. This approach allowed for accuracy calculations but

precluded the use of metrics such as IoU. The future development of evaluation metrics tailored for vector format representations is expected to facilitate more precise evaluations of the produced digital models.

5.2. Future work

(1) Semantic segmentation algorithm for damage detection. The proposed method employed semantic segmentation to segment the structural components. However, it currently does not identify the locations of damages on these components. It is expected that, with the inclusion of data regarding the damages inflicted on the structure, semantic segmentation could be utilized to detect damage, and SSfM could be used to generate a digital model that includes the 3D locations of the damages.

(2) Cable tension prediction. Accurate measurement of cable tension is crucial during the structural health monitoring phase of pedestrian suspension bridges. As time progresses, the tension in the cables may vary, manifesting as changes in the sag ratio. The proposed Hybrid SSfM is capable of accurately measuring this sag ratio. It is anticipated that the cable sag ratios predicted by the proposed method can be utilized for tension prediction, thereby contributing to the structural health monitoring of pedestrian suspension bridges.

(3) FE model updating. The proposed method enabled the digital transformation of pedestrian suspension bridges, including cables. However, the resulting digital models do not incorporate material properties. Consequently, we plan to conduct further research based on updating the finite element (FE) model. By integrating the geometric information from the digital models obtained through the proposed method with vibration data, it is possible to update the FE model. This process is expected to allow for the estimation of critical structural properties, such as stiffness, in structural analysis.

(4) Road and railway suspension bridges. The proposed method is focused on generating digital models of pedestrian suspension bridges, including aspects of cable modeling. The challenges associated with cable modeling are not limited to pedestrian suspension bridges but are also prevalent in road and railway suspension bridges. Due to the more complex and larger scale of road and railway suspension bridges, It is anticipated that utilizing more image data will enable the proposed method to support the digital transformation of not only pedestrian but also road and railway suspension bridges.

6. Conclusion

This paper proposed a technology for the 3D digital transformation of existing pedestrian suspension bridges using deep learning and computer vision, thereby advancing automated inspection and maintenance processes and contributing to Information Technologies in Maintenance. The proposed system consists of three phases: (1) Deck and Pylon Semantic Segmentation, (2) Cable Wireframe Parsing, and (3) 3D Reconstruction and Model Integration. In this process, semantic segmentation and wireframe parsing were combined to represent the semantic information of all structural components, including the crucial cables, using a combination of raster and vector formats. To achieve 3D reconstruction, techniques such as SfM and multiview triangulation were employed. Furthermore, algorithms such as the Thiel–Sen estimator, RANSAC, and K-means clustering were employed to reduce noise and improve model accuracy.

To validate the proposed system, a field experiment was conducted on the Yangban-gil Suspension Bridge located in Goesan, Chungcheongbuk-do, Korea. Performance evaluation for semantic segmentation, including Accuracy and IoU, found that the proposed Hybrid SSfM method outperformed the conventional SSfM approach across all classes. Notably, the IoU results for cable and junction predicted by the conventional SSfM, which utilizes semantic segmentation for all components, were significantly lower at 26.06% and 4.17%, respectively. This is because the cables in the pedestrian suspension bridge images are represented with very thin lines, making it difficult to accurately model their positions using raster format semantic segmentation. The relative error of the sag ratio measured in the 3D cable modeling using the conventional method was 8.6249%, while the proposed method demonstrated a significant improvement with a relative error of 0.4196%. Similarly, the results from SSfM show that the proposed Hybrid SSfM method consistently outperforms the conventional SSfM across the board. Particularly, the accuracy results of SSfM for cable and junction using conventional method were 28.81% and 2.37%, respectively. In contrast, significantly higher accuracies of 93.29% for cables and 65.00% for junctions were exhibited by the model generated using the proposed method. Additionally, while cables generated using the conventional method often suffer from disconnections and some junctions are not detected, the 3D model of the pedestrian suspension bridge created using the proposed method represents the cables in vector format. This ensures continuity and complete junction detection, resulting in significantly improved performance.

Through the method proposed in this study, we successfully generated a digital model of the pedestrian suspension bridge. Practical considerations and limitations of the proposed method were discussed, including deck and pylon modeling, wireframe parsing, UAV flight paths, and the evaluation metrics for vector-format cable representation. Additionally, future research areas were identified, encompassing the semantic segmentation algorithm for damage detection, cable tension prediction, and FE model updating aimed at enhancing the comprehensive structural health monitoring of pedestrian suspension bridges. The feasibility of extending these methodologies to road and railway suspension bridges was also discussed. The contributions of this study to the digital transformation of pedestrian suspension bridges are anticipated to facilitate more cost-effective field inspections in the near future.

CRediT authorship contribution statement

Yeongseo Park: Writing – original draft, Formal analysis, Data curation. **Jaehyuk Lee:** Software, Data curation. **Kevin Han:** Writing – review & editing, Supervision, Resources. **Hyungchul Yoon:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hyungchul Yoon reports financial support was provided by National Research Foundation of Korea. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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